Neurology Review

Epilepsy • Ischemic stroke • Parkinson's disease • Headache • Multiple sclerosis • Dementia — all key points for the exam, with a drug table after each topic

1. Epilepsy

Definitions and classification

- **Seizure:** transient abnormal, excessive neuronal activity • **provoked** (alcohol withdrawal, hypoglycemia, drugs, infection) vs **unprovoked**.
- **Epilepsy: ≥ 2 unprovoked seizures > 24 h apart**, or 1 unprovoked seizure + high (≥ 60%) recurrence risk, or an epilepsy syndrome.
- **Diagnosis:** history (witness), **EEG**, **MRI**, labs (glucose, sodium, calcium, toxicology).

Type (ILAE 2017)	Features
Focal (partial)	Starts in one hemisphere • **focal aware** (simple partial) or **focal impaired awareness** (complex partial) • may spread → **focal to bilateral tonic-clonic**
Generalized tonic-clonic	Loss of consciousness, stiffening (tonic) then jerking (clonic), tongue biting, incontinence, postictal confusion
Absence	**Children** • brief staring spells (seconds), many per day • **3-Hz spike-wave** on EEG
Myoclonic	Brief shock-like jerks (e.g., juvenile myoclonic epilepsy, worse in the morning)
Atonic / tonic	Sudden loss of tone (drop attacks) / stiffening • common in **Lennox-Gastaut syndrome**

Treatment principles

- Usually start after the **second unprovoked seizure** (or after the first if high recurrence risk).
- **Monotherapy** first • **start low, go slow** • titrate to seizure control or side effects • if failure, **switch** to another monotherapy before combining.
- **Narrow-spectrum** drugs (Na-channel blockers, gabapentinoids) can **WORSEN absence and myoclonic** seizures.
- Consider withdrawal after **≥ 2 seizure-free years** → **slow taper** (abrupt stop → seizures/status).
- **All antiseizure drugs:** FDA warning for **suicidal thoughts/behavior** → monitor mood.
- **HLA-B*1502** (Asian ancestry) → ↑ SJS/TEN with **carbamazepine** (also oxcarbazepine, phenytoin/fosphenytoin) → test before starting • **HLA-A*3101** → carbamazepine hypersensitivity.

Drug choice by seizure type

Seizure type	First-line / preferred	Avoid
Focal	**Lamotrigine**, **levetiracetam** • alternatives: oxcarbazepine, carbamazepine, lacosamide, zonisamide, topiramate	—
Generalized tonic-clonic	**Valproate** (most effective; avoid in women of childbearing potential), **lamotrigine**, **levetiracetam**	—
Absence	**Ethosuximide** (first-line in children), valproate, lamotrigine	**Carbamazepine, oxcarbazepine, phenytoin, gabapentin, pregabalin, tiagabine, vigabatrin**
Myoclonic	**Valproate**, **levetiracetam**, topiramate	Same as absence • **lamotrigine may worsen**
Lennox-Gastaut	Valproate, lamotrigine, **rufinamide**, **clobazam**, cannabidiol, felbamate, fenfluramine	—
Dravet syndrome	Valproate, clobazam, stiripentol, cannabidiol, **fenfluramine**	**Na-channel blockers**
Infantile spasms	**ACTH**, prednisolone, **vigabatrin** (tuberous sclerosis)	—

Therapeutic drug monitoring

Drug	Target level	Key point
Phenytoin	**10–20 mcg/mL** (free 1–2)	**Non-linear (Michaelis-Menten)** kinetics → small dose ↑ = large level ↑ • **correct for low albumin**: corrected = measured ÷ [(0.2 × albumin) + 0.1] (use 0.1 × albumin + 0.1 if CrCl < 10)
Carbamazepine	**4–12 mcg/mL**	**Autoinduction** (levels fall over 2–4 weeks → dose ↑)
Valproate	**50–100 mcg/mL** (epilepsy)	Highly protein-bound → check **free level** in hypoalbuminemia
Phenobarbital	15–40 mcg/mL	Very long half-life
Ethosuximide	40–100 mcg/mL	

Key drug facts

Drug	Mechanism	Key ADRs / boxed warnings	Key DDIs / notes
Phenytoin	Na-channel blocker	**Dose-related:** nystagmus (> 20), ataxia (> 30), confusion (> 40) • **Chronic: gingival hyperplasia**, **hirsutism**, coarse facies, **osteomalacia** (↓ vitamin D), **folate deficiency** • SJS • **IV:** hypotension, arrhythmias (max **50 mg/min**), purple glove syndrome • teratogen	**Strong inducer** (3A4, 2C9, 2C19, UGT) → ↓ OCs, many drugs • ↑ levels: **fluconazole**, amiodarone, isoniazid, valproate (displacement) • **enteral feeds** ↓ absorption (hold 1–2 h)
Fosphenytoin	Prodrug of phenytoin	Less infusion-site toxicity • max **150 mg PE/min** • hypotension, pruritus	Dosed in **phenytoin equivalents (PE)**
Carbamazepine	Na-channel blocker	**Boxed: SJS/TEN** (HLA-B*1502), **aplastic anemia/agranulocytosis** • **hyponatremia (SIADH)**, dizziness, diplopia, ataxia, hepatotoxicity • teratogen (neural tube)	**Strong inducer + autoinducer** • 3A4 substrate: ↑ by **macrolides**, **azoles**, grapefruit, diltiazem/verapamil • **CI with MAOIs**, nefazodone
Oxcarbazepine	Na-channel blocker	**Hyponatremia** (more than CBZ), dizziness, SJS • 25–30% cross-reactivity with CBZ	Weaker inducer (↓ OCs at higher doses)
Valproate (divalproex)	Broad (Na, Ca, ↑ GABA)	**Boxed: hepatotoxicity** (children < 2), **pancreatitis**, **teratogenicity** (neural tube defects, ↓ IQ) • weight gain, tremor, **alopecia**, **thrombocytopenia**, **hyperammonemia** (L-carnitine), PCOS	**Enzyme inhibitor** → ↑ **lamotrigine** (halve LTG dose), ↑ phenobarbital • **carbapenems ↓ valproate** markedly (avoid) • + topiramate → hyperammonemia
Lamotrigine	Na-channel blocker	**Boxed: serious rash (SJS/TEN)** — risk ↑ with **fast titration**, **valproate**, children • dizziness, diplopia, HLH (rare)	**Slow titration** • **valproate** ↑ LTG (use half dose) • **inducers** ↓ (double dose) • **estrogen OCs ↓ LTG ~50%** • levels fall in **pregnancy** • restart titration if missed > 5 half-lives
Levetiracetam	SV2A binding	**Behavioral changes** (irritability, aggression, depression, psychosis), somnolence, fatigue	**Few DDIs** • **renal dose adjustment** • IV form • preferred in elderly and pregnancy
Topiramate	Broad	**Cognitive slowing**, word-finding difficulty, **paresthesias**, **kidney stones**, **metabolic acidosis**, **weight loss**, **oligohidrosis**/hyperthermia (children), **acute angle-closure glaucoma** • teratogen (cleft lip)	↓ OCs at > 200 mg/day • also migraine prevention
Zonisamide	Na/Ca (sulfonamide)	Kidney stones, oligohidrosis, metabolic acidosis, weight loss • sulfa allergy	Long half-life (daily)
Lacosamide	Slow inactivation of Na channels	Dizziness, **PR prolongation** (ECG in cardiac disease)	IV form • Schedule V
Gabapentin / pregabalin	α2δ calcium channel	Dizziness, **sedation**, weight gain, **edema**, misuse	Focal only • **renal adjustment** • can worsen absence/myoclonic
Phenobarbital / primidone	GABA-A (barbiturate)	Sedation, cognitive and behavioral effects, dependence	**Strong inducer**
Ethosuximide	T-type Ca-channel blocker	GI upset, hiccups, drowsiness, rash/SJS	**Absence only**
Clobazam	Benzodiazepine	Sedation, SJS, dependence	Lennox-Gastaut • ↑ by cannabidiol
Cenobamate	Na + GABA	**DRESS** (slow titration), **QT shortening** (CI familial short QT)	↓ OCs
Perampanel	AMPA antagonist	**Boxed: serious psychiatric/behavioral reactions** (aggression, homicidal ideation)	Bedtime dosing

Drug	Mechanism	Key ADRs / boxed warnings	Key DDIs / notes
Felbamate	NMDA + GABA	**Boxed: aplastic anemia, hepatic failure**	Refractory only
Vigabatrin	GABA-transaminase inhibitor	**Boxed: permanent vision loss** (REMS, eye exams)	Infantile spasms
Cannabidiol (Epidiolex)	Unknown	↑ LFTs, somnolence, diarrhea	↑ clobazam active metabolite • check LFTs
Fenfluramine	Serotonergic	**Valvular heart disease**, pulmonary hypertension (echo; REMS)	Dravet, LGS

Women of childbearing potential and pregnancy

- **Avoid valproate** (highest risk: neural tube defects, ↓ IQ, autism) • also teratogenic: carbamazepine, phenytoin, phenobarbital, topiramate.
- **Preferred: lamotrigine** and **levetiracetam**.
- **Folic acid 0.4–1 mg daily** (higher doses, e.g., 4 mg, in some guidelines) **before conception** and in pregnancy.
- **Enzyme inducers** (carbamazepine, phenytoin, phenobarbital, oxcarbazepine, topiramate > 200 mg, cenobamate, perampanel) ↓ hormonal contraceptives → use an **IUD** or depot medroxyprogesterone.
- **Lamotrigine levels drop** during pregnancy and with estrogen → monitor levels and adjust.
- Most antiseizure drugs are compatible with breastfeeding.

Other special populations

- **Elderly:** levetiracetam, lamotrigine preferred • avoid enzyme inducers (DDIs, bone loss) • watch hyponatremia (CBZ, OXC) and falls.
- **Bone health:** enzyme inducers ↓ vitamin D → calcium + vitamin D supplementation.
- **Drugs that lower seizure threshold: bupropion**, **tramadol**, meperidine, **imipenem**, cefepime (renal failure), fluoroquinolones, **theophylline**, clozapine, lithium toxicity, **isoniazid** (treat with **pyridoxine**), flumazenil in benzodiazepine users.

Status epilepticus (seizure ≥ 5 minutes, or repeated without recovery)

Phase	Treatment
0–5 min: stabilize	ABCs, oxygen, IV access • check **glucose** → if low: **thiamine 100 mg IV then dextrose** (adults)
5–20 min: benzodiazepine (first-line)	**Lorazepam IV 0.1 mg/kg** (max 4 mg/dose, may repeat once) **OR midazolam IM 10 mg** (> 40 kg; 5 mg if 13–40 kg) **OR diazepam IV 0.15–0.2 mg/kg** (max 10 mg) • pre-hospital: **rectal diazepam**, **intranasal/buccal midazolam**
20–40 min: second line (one of)	**Fosphenytoin 20 mg PE/kg IV** (max 1,500 mg PE) • **valproate 40 mg/kg IV** (max 3,000 mg) • **levetiracetam 60 mg/kg IV** (max 4,500 mg) — equally effective (ESETT) • phenobarbital 15 mg/kg if none available
> 40 min: refractory	ICU: **continuous infusion** of **midazolam**, **propofol**, or **pentobarbital** (± ketamine) with EEG monitoring

- **Home rescue medications** for seizure clusters: **diazepam rectal gel** (Diastat), **intranasal midazolam** (Nayzilam 5 mg), **intranasal diazepam** (Valtoco).

Epilepsy drug summary

Drug	Usual adult dose	Key ADRs	Key DDIs / notes
Levetiracetam	500 mg BID → max 1,500 mg BID	Irritability, aggression, depression, somnolence	Renal adjustment • few DDIs
Lamotrigine	25 mg daily × 2 wk → 50 mg × 2 wk → ↑ 50 mg q1–2 wk; maintenance 225–375 mg/day (with VPA: 25 mg every other day start; 100–200 mg/day)	**Rash/SJS**, dizziness, diplopia	VPA ↑ (halve) • inducers ↓ • estrogen ↓
Valproate	10–15 mg/kg/day → max 60 mg/kg/day	**Hepatotoxicity, pancreatitis, teratogen**, weight gain, tremor, alopecia, ↓ platelets, ↑ NH3	Inhibitor (↑ LTG, PB) • carbapenems ↓ VPA
Carbamazepine	200 mg BID → 800–1,200 mg/day	**SJS (HLA-B*1502)**, aplastic anemia, **hyponatremia**	Strong inducer, autoinducer • macrolides/azoles ↑
Oxcarbazepine	300 mg BID → 1,200–2,400 mg/day	**Hyponatremia**, dizziness	↓ OCs
Phenytoin	Load 15–20 mg/kg; maintenance 300–400 mg/day (4–7 mg/kg)	Gingival hyperplasia, hirsutism, ataxia, nystagmus	Strong inducer • albumin correction • tube feeds
Fosphenytoin	Load 20 mg PE/kg IV (max 150 mg PE/min)	Hypotension, arrhythmia	PE units
Topiramate	25–50 mg/day → 200–400 mg/day	Cognitive slowing, stones, acidosis, glaucoma	↓ OCs (> 200 mg)
Zonisamide	100 mg daily → 400 mg/day	Stones, oligohidrosis	Sulfa allergy
Lacosamide	50–100 mg BID → 200–400 mg/day	Dizziness, PR prolongation	ECG in cardiac disease
Ethosuximide	250 mg BID → 20–40 mg/kg/day	GI, hiccups	Absence only
Gabapentin	300 mg TID → 1,800–3,600 mg/day	Sedation, edema	Renal adjustment
Pregabalin	75 mg BID → 600 mg/day	Dizziness, edema, weight gain	Renal adjustment • Schedule V
Phenobarbital	1–3 mg/kg/day	Sedation, dependence	Strong inducer
Lorazepam IV	0.1 mg/kg (max 4 mg/dose) for status	Respiratory depression, hypotension	First-line for status
Midazolam IM / intranasal	10 mg IM (> 40 kg) • nasal 5 mg	Respiratory depression	No IV access needed
Diazepam rectal / nasal	0.2–0.5 mg/kg rectal (age-based) • nasal 5–20 mg	Sedation	Home rescue

2. Ischemic Stroke

Overview

- **Ischemic stroke ~87%** (thrombotic, embolic — e.g., from **atrial fibrillation**) vs hemorrhagic (intracerebral, subarachnoid).
- **TIA:** transient symptoms without infarction — high early stroke risk (ABCD2 score).
- **Recognize: BE-FAST** — Balance, Eyes, Face drooping, Arm weakness, Speech difficulty, Time to call emergency.

- **First step: non-contrast head CT → rule out hemorrhage** before any thrombolytic or antithrombotic • severity by **NIHSS**.
- **Risk factors: hypertension** (most important), AF, diabetes, dyslipidemia, smoking, obesity, inactivity, prior stroke/TIA, carotid stenosis.

Acute treatment: thrombolysis

Point	Details
Drugs	**Alteplase 0.9 mg/kg IV** (max **90 mg**): **10% as a bolus over 1 minute**, remaining **90% over 60 minutes** • **Tenecteplase 0.25 mg/kg IV single bolus** (max **25 mg**) — now an approved alternative
Time window	Within **3 hours** of symptom onset/last known well • **up to 4.5 hours** in selected patients
BP requirement	Must be **< 185/110 mmHg before** and **< 180/105 mmHg for 24 hours after** → **labetalol** 10–20 mg IV, **nicardipine** 5 mg/h IV (titrate), or clevidipine
Key exclusions	Intracranial hemorrhage on CT • ischemic stroke or severe head trauma in past **3 months** • intracranial/spinal surgery in 3 months • GI bleeding within 21 days • active bleeding • **platelets < 100,000** • **INR > 1.7** or ↑ aPTT • full-dose **LMWH within 24 h** • **DOAC within 48 h** (unless labs normal) • **glucose < 50 mg/dL** • BP uncontrolled > 185/110 • aortic dissection • intracranial tumor/AVM
After thrombolysis	**No antiplatelets or anticoagulants for 24 hours** • neuro checks and BP monitoring • watch for **ICH** and **angioedema** (higher with ACE inhibitors)
Mechanical thrombectomy	For **large vessel occlusion** — up to **24 hours** in selected patients (can follow tPA)

Acute treatment: patients not receiving thrombolysis

- **Permissive hypertension:** treat only if **> 220/120 mmHg** (lower by ~15% in first 24 h) in the first 48–72 hours.
- **Aspirin 160–325 mg within 24–48 hours** (delay to 24 h after tPA).
- **Dual antiplatelet therapy (DAPT)** for **minor stroke (NIHSS ≤ 3)** or **high-risk TIA (ABCD2 ≥ 4)**: **aspirin + clopidogrel** (clopidogrel load 300–600 mg) for **21 days**, then single antiplatelet • or **aspirin + ticagrelor** for 30 days.
- **Glucose 140–180 mg/dL** • treat fever • **VTE prophylaxis** (IPC devices; LMWH/UFH) • swallow screen before oral intake • start **high-intensity statin**.

Secondary prevention

Situation	Therapy
Non-cardioembolic stroke/TIA	**Single antiplatelet long-term**: **aspirin 81–325 mg**, **clopidogrel 75 mg**, or **aspirin/extended-release dipyridamole 25/200 mg BID** (headache) • **no long-term DAPT** (bleeding) except special cases (e.g., intracranial stenosis for 90 days)
Atrial fibrillation	**Oral anticoagulant** — **DOAC preferred** over warfarin (apixaban, rivaroxaban, edoxaban, dabigatran) • start **~4–14 days** after stroke depending on size (earlier for small strokes) • not aspirin instead
Lipids	**High-intensity statin** (atorvastatin 80 mg) • target **LDL < 70 mg/dL** (add ezetimibe/PCSK9i)
Blood pressure	Target **< 130/80 mmHg** • thiazide, ACEi/ARB, CCB
Other	Diabetes control • **smoking cessation** • weight, exercise, Mediterranean diet, limit alcohol • **carotid endarterectomy/stenting** for symptomatic carotid stenosis 70–99% • PFO closure (selected) • avoid estrogen-containing therapy

Hemorrhagic stroke (brief)

- **Intracerebral hemorrhage:** lower **SBP to ~140 mmHg** (130–150) • **reverse anticoagulation**: warfarin → **4-factor PCC + IV vitamin K** • dabigatran → **idarucizumab** • factor Xa inhibitors → **andexanet alfa** or 4F-PCC • no routine seizure prophylaxis.

- **Subarachnoid hemorrhage:** "worst headache of my life" • **nimodipine 60 mg PO every 4 hours × 21 days** (prevents delayed ischemia from vasospasm; hypotension → 30 mg q2h).

Stroke drug summary

Drug	Dose	Key ADRs	Key DDIs / notes
Alteplase	0.9 mg/kg (max 90 mg): 10% bolus, 90% over 60 min	**ICH**, bleeding, **angioedema**	BP < 185/110 before; no antithrombotics × 24 h • ACEi ↑ angioedema
Tenecteplase	0.25 mg/kg IV bolus (max 25 mg)	ICH, bleeding	Single bolus
Labetalol	10–20 mg IV over 1–2 min, may repeat	Bradycardia, hypotension, bronchospasm	Avoid in asthma, heart block
Nicardipine	5 mg/h IV, ↑ 2.5 mg/h q5–15 min (max 15)	Hypotension, reflex tachycardia, headache	Titratable
Aspirin	Acute 160–325 mg • prevention 81–325 mg daily	GI bleeding	↑ bleeding with anticoagulants, NSAIDs
Clopidogrel	Load 300–600 mg → 75 mg daily	Bleeding, TTP (rare)	**Omeprazole/esomeprazole** ↓ effect (CYP2C19) → use pantoprazole • CYP2C19 poor metabolizers
Ticagrelor	180 mg load → 90 mg BID (with aspirin ≤ 100 mg)	Bleeding, **dyspnea**	Strong CYP3A4 inhibitors/inducers avoid
Aspirin/ER dipyridamole	25/200 mg BID	**Headache**, GI	Do not substitute IR dipyridamole
Apixaban	5 mg BID (2.5 mg if ≥ 2 of: age ≥ 80, weight ≤ 60 kg, SCr ≥ 1.5)	Bleeding	Strong CYP3A4/P-gp inducers avoid
Rivaroxaban	20 mg daily with evening meal (15 mg if CrCl 15–50)	Bleeding	Take with food • CYP3A4/P-gp
Dabigatran	150 mg BID (CrCl > 30)	Bleeding, **dyspepsia**	P-gp • keep in original bottle • reversal idarucizumab
Warfarin	Dose to INR 2–3	Bleeding, skin necrosis	Many DDIs (CYP2C9) • vitamin K diet consistency
Atorvastatin	40–80 mg daily	Myalgia, ↑ LFTs	CYP3A4 inhibitors ↑
Nimodipine	60 mg PO q4h × 21 days (SAH)	Hypotension	**Never IV** (fatal) • CYP3A4
4-factor PCC	Weight/INR-based (25–50 units/kg)	Thrombosis	Warfarin (with vitamin K 10 mg IV) or Xa-inhibitor reversal
Idarucizumab	5 g IV	Thrombosis	Dabigatran reversal
Andexanet alfa	Bolus + infusion (dose by agent/time)	Thrombosis (boxed)	Apixaban/rivaroxaban reversal

3. Parkinson's Disease (PD)

Overview

- Loss of **dopaminergic neurons** in the **substantia nigra** → ↓ dopamine and relative ↑ acetylcholine in the striatum.
- **Motor features (TRAP): T**remor (resting, "pill-rolling"), **R**igidity (cogwheel), **A**kinesia/**bradykinesia** (most disabling), **P**ostural instability (late) • shuffling gait, masked face, micrographia.
- **Non-motor: depression**, anxiety, **constipation**, **orthostatic hypotension**, **REM sleep behavior disorder**, anosmia, **dementia**, **psychosis**, sialorrhea, urinary problems.
- **Drug-induced parkinsonism: antipsychotics** (especially FGAs), **metoclopramide**, prochlorperazine, promethazine, reserpine, tetrabenazine, lithium, valproate.

Treatment approach

- Start treatment when symptoms **affect function or quality of life** • exercise and physical therapy for all.
- **Carbidopa/levodopa** = **most effective** → first-line for most patients (especially **older** or significant disability).
- **Dopamine agonists** → option in **younger** patients (< 65) to delay motor complications (but more psychiatric/impulse-control side effects).
- **MAO-B inhibitors** → mild early symptoms (modest effect).
- **Amantadine** → mild symptoms or **levodopa-induced dyskinesia** • **Anticholinergics** → **tremor** in young patients only.

Carbidopa/levodopa

- **Levodopa** crosses the blood-brain barrier → converted to dopamine • **carbidopa** blocks **peripheral dopa decarboxylase** → more levodopa reaches the brain and ↓ nausea (need ≥ 75 mg carbidopa/day).
- **Start:** 25/100 mg **TID** • titrate.
- **ADRs: nausea**, **orthostatic hypotension**, **dyskinesias**, **hallucinations/psychosis**, confusion, impulse-control disorders, dark urine/sweat.
- **Interactions: high-protein meals** compete for absorption (spread protein across the day) • **iron** ↓ absorption • **non-selective MAOIs** → hypertensive crisis (**stop MAOI 14 days before**) • **dopamine antagonists** (antipsychotics, metoclopramide) oppose effect • pyridoxine (only with levodopa alone).
- **Never stop abruptly** → risk of neuroleptic malignant-like syndrome.

Motor complications (after years of levodopa)

Problem	Management
"Wearing-off" (end-of-dose)	**More frequent levodopa doses** • add **COMT inhibitor** (entacapone, opicapone), **MAO-B inhibitor**, **dopamine agonist**, or **istradefylline** • extended-release levodopa (Rytary)
"On-off" (sudden, unpredictable)	**Rescue**: **apomorphine SC** (with antiemetic **trimethobenzamide**; **never with ondansetron/5-HT3 antagonists** → severe hypotension), **inhaled levodopa** (Inbrija), sublingual apomorphine • continuous infusions (levodopa enteral gel, subcutaneous foscarbidopa/foslevodopa) • deep brain stimulation
Peak-dose dyskinesia	**Reduce levodopa dose** (more frequent, smaller doses) • add **amantadine** (ER: Gocovri)
Freezing	Physical therapy, cues; adjust medications

Other antiparkinson drugs

Class / drug	Key points
Dopamine agonists: **pramipexole**, **ropinirole**, **rotigotine** (patch), apomorphine	ADRs: **nausea**, **orthostatic hypotension**, **hallucinations**, **sudden sleep attacks**/somnolence (driving), **impulse-control disorders** (gambling, shopping, hypersexuality, binge eating), **peripheral edema** • pramipexole: **renal adjustment** • ropinirole: **CYP1A2** (ciprofloxacin, fluvoxamine ↑; smoking ↓) • rotigotine: sulfite allergy, rotate patch sites • also used for **restless legs**
MAO-B inhibitors: **selegiline**, **rasagiline**, safinamide	Selegiline → **amphetamine metabolites** → **insomnia** (dose with breakfast and lunch) • DDIs: **serotonin syndrome** with **meperidine (CI)**, tramadol, methadone, **dextromethorphan**, St. John's wort; caution with SSRIs/SNRIs/TCAs • sympathomimetics • **ciprofloxacin ↑ rasagiline** (max 0.5 mg)
COMT inhibitors: **entacapone**, **opicapone**, tolcapone	Prolong levodopa effect (wearing-off) • **only with levodopa** • entacapone: **brown-orange urine**, **diarrhea**, ↑ dyskinesia • **tolcapone: boxed hepatotoxicity** (LFT monitoring) • Stalevo = carbidopa/levodopa/entacapone
Amantadine	NMDA antagonist • mild PD, **dyskinesia** • ADRs: **livedo reticularis**, **ankle edema**, hallucinations, confusion, anticholinergic effects • **renal adjustment**
Anticholinergics: **trihexyphenidyl**, **benztropine**	**Tremor** in **young** patients • **avoid in elderly** (Beers: confusion, urinary retention, constipation, dry mouth, glaucoma)
Istradefylline	Adenosine A2A antagonist • adjunct for "off" episodes • dyskinesia, nausea • CYP3A4

Non-motor symptoms

Symptom	Treatment
Psychosis (hallucinations)	First: **reduce/stop** anticholinergics → amantadine → dopamine agonists → MAO-B/COMT • then **pimavanserin** (34 mg daily; 5-HT2A inverse agonist; QT; boxed elderly dementia mortality) or low-dose **quetiapine**/**clozapine** • **avoid typical antipsychotics, risperidone, olanzapine** (worsen motor symptoms)
Dementia	**Rivastigmine** (FDA-approved for PD dementia) • donepezil
Depression / anxiety	SSRIs, SNRIs (watch serotonin syndrome with MAO-B inhibitors)
Orthostatic hypotension	Fluids, salt, compression • **fludrocortisone**, **midodrine**, **droxidopa** (boxed: supine hypertension)
Constipation	Fiber, fluids, PEG
Sialorrhea	Botulinum toxin, glycopyrrolate
REM sleep behavior disorder	**Melatonin**, clonazepam

Parkinson's drug summary

Drug	Dose	Key ADRs	Key DDIs / notes
Carbidopa/levodopa	25/100 mg TID → titrate (ER and CR forms)	Nausea, orthostasis, **dyskinesia**, hallucinations	Protein meals, iron ↓ • **non-selective MAOIs CI** • D2 antagonists oppose
Inhaled levodopa (Inbrija)	84 mg PRN (up to 5×/day)	Cough, URI	Rescue for "off" • not in asthma/COPD
Pramipexole	0.125 mg TID → 0.5–1.5 mg TID	Sleep attacks, ICDs, hallucinations, edema	**Renal adjustment**
Ropinirole	0.25 mg TID → max 24 mg/day	Same	**CYP1A2** (cipro ↑, smoking ↓)
Rotigotine patch	2 mg/24 h → 6–8 mg/24 h	Application-site reactions + agonist effects	Sulfite allergy • remove before MRI
Apomorphine SC	2 mg test dose → 2–6 mg PRN	Severe nausea, hypotension, injection site	**CI with 5-HT3 antagonists** • pretreat with trimethobenzamide
Selegiline	5 mg BID (breakfast, lunch)	**Insomnia**, nausea	**Meperidine CI** • serotonergic drugs • tyramine at high doses
Rasagiline	0.5–1 mg daily	Headache, arthralgia	Cipro ↑ (max 0.5 mg) • serotonergic drugs
Safinamide	50 → 100 mg daily	Dyskinesia	Adjunct to levodopa • serotonergic drugs
Entacapone	200 mg with each levodopa dose (max 8/day)	**Brown-orange urine**, **diarrhea**, dyskinesia	Only with levodopa
Opicapone	50 mg at bedtime	Dyskinesia, constipation	Once daily COMT inhibitor
Tolcapone	100 mg TID	**Boxed: hepatotoxicity**	LFT monitoring
Amantadine	100 mg BID (ER 137–274 mg HS)	**Livedo reticularis**, edema, hallucinations	**Renal adjustment**
Trihexyphenidyl	1 mg → 2 mg TID	Anticholinergic	Avoid elderly
Benztropine	0.5–2 mg daily–BID	Anticholinergic	Avoid elderly
Istradefylline	20–40 mg daily	Dyskinesia, nausea	CYP3A4; smokers need 40 mg
Pimavanserin	34 mg daily	Edema, confusion, **QT**	**Boxed: ↑ mortality in elderly with dementia** • CYP3A4
Droxidopa	100 mg TID → max 600 mg TID	**Boxed: supine hypertension**	Last dose ≥ 3 h before bedtime

4. Headache

Red flags for secondary headache (SNOOP)

Systemic symptoms (fever, weight loss, cancer, HIV) • **N**eurologic signs • **O**nset sudden ("**thunderclap**" → SAH) • **O**lder age (> 50, new headache → giant cell arteritis) • **P**attern change, positional, papilledema, pregnancy → **urgent evaluation**.

Primary headache types

Feature	Migraine	Tension-type	Cluster
Location	Usually **unilateral**	**Bilateral** (band-like)	**Unilateral, orbital/temporal**
Quality	**Pulsating**	Pressing/tightening	Severe, sharp, boring
Severity	Moderate–severe, worse with activity	Mild–moderate	**Very severe**
Duration	**4–72 hours**	30 min–7 days	**15–180 min**, up to 8/day, in clusters (weeks)
Associated	**Nausea/vomiting, photophobia, phonophobia** • aura (~25%)	None (no nausea)	**Autonomic**: tearing, rhinorrhea, ptosis, miosis • **restlessness**
Who	Women > men (3:1)	Most common overall	**Men**, smokers, circadian timing

Migraine: acute treatment

- **Treat early** in the attack • **stratified care** by severity.
- **Mild–moderate: NSAIDs** (ibuprofen 400 mg, naproxen 500–825 mg, aspirin 900–1,000 mg, diclofenac potassium 50 mg), **acetaminophen 1,000 mg**, or **acetaminophen/aspirin/caffeine**.
- **Moderate–severe** (or NSAID failure): **triptans** (± NSAID; e.g., sumatriptan/naproxen) • **gepants** or **lasmiditan** if triptans fail or are **contraindicated** (cardiovascular disease).
- **Nausea/vomiting:** non-oral routes (SC/nasal triptan) + **antiemetic** (metoclopramide, prochlorperazine).
- **Emergency department:** IV **metoclopramide 10 mg** or **prochlorperazine** (+ diphenhydramine to prevent akathisia), IV ketorolac, IV DHE • **dexamethasone 10 mg IV** ↓ recurrence • **avoid opioids and butalbital**.

Class	Key points
Triptans (5-HT1B/1D agonists)	**Sumatriptan** 50–100 mg PO (6 mg SC, 20 mg nasal), **rizatriptan** 10 mg (**5 mg with propranolol**), zolmitriptan, eletriptan (CYP3A4), almotriptan, **naratriptan** and **frovatriptan** (long half-life, slower) • may **repeat after 2 hours** (max sumatriptan 200 mg/day) • **Contraindicated: CAD/ischemic heart disease**, prior **stroke/TIA**, **uncontrolled hypertension**, PVD, **hemiplegic/basilar migraine**, within **24 h of ergots or another triptan**, **MAOIs** (sumatriptan, rizatriptan, zolmitriptan) • ADRs: chest/neck tightness, tingling, flushing, dizziness • caution serotonin syndrome with SSRIs/SNRIs
Gepants (CGRP receptor antagonists)	**Ubrogepant** 50–100 mg, **rimegepant** 75 mg ODT (also prevention), **zavegepant** 10 mg nasal • **no vasoconstriction** → OK in cardiovascular disease • no medication-overuse headache • ubrogepant/rimegepant: avoid **strong CYP3A4 inhibitors**
Ditan	**Lasmiditan** 50–200 mg (5-HT1F agonist) • no vasoconstriction • **dizziness, sedation** → **no driving for 8 hours** • Schedule V
Ergots	**Dihydroergotamine** (DHE nasal/IV/SC), ergotamine • **CI: CV disease**, **pregnancy**, **strong CYP3A4 inhibitors** (macrolides, azole antifungals, protease inhibitors → **ergotism**), within 24 h of triptans

Medication overuse headache (MOH)

- Headache ≥ 15 days/month with regular overuse for > 3 months: **simple analgesics ≥ 15 days/month**, or **triptans, ergots, opioids, combination analgesics ≥ 10 days/month**.
- **Prevention:** limit acute treatment to **≤ 2–3 days per week** • **Treatment:** withdraw the overused drug (taper opioids/butalbital) and start preventive therapy.

Migraine: preventive treatment

- **Indications: ≥ 4 headache days/month** (or ≥ 3 with significant disability), acute therapy failure/overuse/contraindication, patient preference, hemiplegic migraine or prolonged aura.
- Start low, titrate • adequate trial **2–3 months** • goal ≥ **50% reduction** • continue 6–12 months, then consider taper.

- Choose based on **comorbidities** (e.g., propranolol for hypertension; amitriptyline for insomnia/depression; topiramate if overweight).

Agent	Dose	Notes
Propranolol	80–240 mg/day	Also metoprolol, timolol • avoid in asthma • fatigue, depression
Topiramate	25 mg → 50 mg BID	Cognitive effects, paresthesias, weight loss, stones • teratogen • ↓ OCs
Divalproex	500–1,000 mg/day	**Avoid in women of childbearing potential** • weight gain, tremor
CGRP monoclonal antibodies	**Erenumab** 70–140 mg SC monthly • **fremanezumab** 225 mg monthly or 675 mg quarterly • **galcanezumab** 240 mg load → 120 mg monthly • **eptinezumab** 100 mg IV every 3 months	Injection site reactions • **erenumab: constipation, hypertension**
Gepants (preventive)	**Atogepant** 10–60 mg daily • **rimegepant** 75 mg every other day	Nausea, constipation • CYP3A4 DDIs
OnabotulinumtoxinA	155 units IM every 12 weeks	**Chronic migraine only** (≥ 15 headache days/month)
Amitriptyline	10–25 mg HS → up to 150 mg	Anticholinergic, sedation, weight gain
Venlafaxine	75–150 mg/day	↑ BP
Candesartan	16 mg/day	Hypotension, hyperkalemia
Supplements	Magnesium 400–600 mg, **riboflavin 400 mg**, CoQ10, feverfew	Butterbur — avoid (hepatotoxicity)

Special situations

- **Migraine with aura: avoid estrogen-containing contraceptives** (↑ ischemic stroke risk) → progestin-only or non-hormonal.
- **Menstrual migraine:** short-term prevention with **frovatriptan** or **naratriptan** (or NSAID) around menses.
- **Pregnancy: acetaminophen** first • metoclopramide • sumatriptan if needed • **avoid valproate, topiramate, ergots**, NSAIDs in 3rd trimester • prevention: propranolol.
- **Children:** ibuprofen, acetaminophen • triptans approved: **rizatriptan (≥ 6 years)**, almotriptan, zolmitriptan nasal, sumatriptan/naproxen (≥ 12).

Tension-type and cluster headache

- **Tension-type acute: acetaminophen**, **NSAIDs**, aspirin (± caffeine) • **chronic prevention: amitriptyline** • avoid MOH.
- **Cluster acute: 100% oxygen 12–15 L/min** via non-rebreather mask for 15 minutes • **sumatriptan 6 mg SC** • zolmitriptan nasal.
- **Cluster prevention: verapamil** 240–960 mg/day (first-line; **ECG** — heart block) • **bridge**: prednisone short course or occipital nerve block • lithium • **galcanezumab 300 mg monthly** (episodic cluster).

Headache drug summary

Drug	Dose	Key ADRs	Key DDIs / notes
Ibuprofen / naproxen	400–800 mg / 500–825 mg at onset	GI bleed, renal	Limit use (MOH)
Acetaminophen	1,000 mg at onset (max 3–4 g/day)	Hepatotoxicity (overdose)	First-line in pregnancy
Sumatriptan	50–100 mg PO, may repeat in 2 h (max 200 mg/day) • 6 mg SC (max 12 mg) • 20 mg nasal	Chest tightness, tingling	**CI: CAD, stroke, uncontrolled HTN, MAOIs**, ergots within 24 h
Rizatriptan	10 mg (5 mg with propranolol), max 30 mg/day	Same	**Propranolol ↑ levels** • MAOIs CI
Eletriptan	20–40 mg (max 80 mg/day)	Same	**Strong CYP3A4 inhibitors CI**
Frovatriptan / naratriptan	2.5 mg (repeat in 2 h / 4 h)	Same (milder)	Long half-life • menstrual migraine
Ubrogepant	50–100 mg, may repeat in 2 h (max 200 mg/day)	Nausea, somnolence	Strong CYP3A4 inhibitors CI
Rimegepant	75 mg ODT (acute) • 75 mg every other day (prevention)	Nausea	Strong CYP3A4 inhibitors avoid
Zavegepant	10 mg nasal (max 1/day)	Taste disturbance, nasal discomfort	—
Lasmiditan	50–200 mg (max 1 dose/24 h)	**Dizziness, sedation**	**No driving 8 h** • Schedule V
Dihydroergotamine	1 mg IV/IM/SC • 2 mg nasal	Nausea, vasospasm	**CI: CV disease, pregnancy, strong CYP3A4 inhibitors**, triptans within 24 h
Metoclopramide	10 mg IV/PO	**Akathisia**, dystonia, TD (boxed)	Give diphenhydramine
Prochlorperazine	10 mg IV/IM	Akathisia, sedation, QT	Dopamine antagonist
Propranolol	80–240 mg/day (prevention)	Fatigue, bradycardia	Avoid asthma • ↑ rizatriptan
Topiramate	50 mg BID (prevention)	Cognitive slowing, stones	Teratogen • ↓ OCs
Erenumab	70–140 mg SC monthly	**Constipation**, **hypertension**	Latex in device (some)
Fremanezumab / galcanezumab / eptinezumab	See table above	Injection site reactions	—
Atogepant	10–60 mg daily	Nausea, constipation	Strong CYP3A4 inhibitors → 10 mg
Onabotulinumtoxin A	155 units q12 wk	Neck pain, ptosis	Chronic migraine
Amitriptyline	10–150 mg HS	Anticholinergic, sedation	Beers
Verapamil (cluster)	240–960 mg/day	**Bradycardia, heart block**, constipation	ECG • CYP3A4 inhibitor
Oxygen (cluster)	100% at 12–15 L/min × 15 min	—	First-line acute cluster

5. Multiple Sclerosis (MS)

Overview

- **Autoimmune demyelination** of the CNS (brain, spinal cord, optic nerves) • more common in **women**, onset **20–40 years**.
- **Types: clinically isolated syndrome (CIS) • relapsing-remitting (RRMS, ~85%) • secondary progressive (SPMS) • primary progressive (PPMS)**.
- **Symptoms: optic neuritis** (painful vision loss), numbness/paresthesias, weakness, **spasticity**, **fatigue**, bladder and bowel dysfunction, ataxia, depression, cognitive problems • **Uhthoff phenomenon** (worse with heat) • **Lhermitte sign** (electric shock down spine when bending the neck).
- **Diagnosis (McDonald criteria):** lesions disseminated in **time and space** (MRI) • CSF **oligoclonal bands**.

Acute relapse

- **High-dose corticosteroids: methylprednisolone 1 g IV daily × 3–5 days** (or equivalent high-dose oral prednisone) ± short taper → speeds recovery (does not change long-term disability).
- Refractory severe relapse → **plasma exchange** • ACTH gel (alternative).

Disease-modifying therapies (DMTs)

- Start **early** in **relapsing MS** (and CIS with MRI lesions) → ↓ relapses, new MRI lesions, disability progression • **do not reverse** damage.
- **PPMS: ocrelizumab** is the only approved DMT • active SPMS: siponimod and other relapsing-MS DMTs.
- Before high-efficacy DMTs: screen **hepatitis B**, TB, **VZV immunity** (vaccinate), JCV antibodies (natalizumab), CBC, LFTs • give **live vaccines before** starting.

DMT	Dose	Key ADRs / monitoring
Interferon beta-1a (Avonex IM weekly; Rebif SC 3×/week; peginterferon Plegridy SC every 2 weeks) • **beta-1b** (Betaseron SC every other day)	Avonex 30 mcg IM weekly • Rebif 22–44 mcg SC TIW • Betaseron 0.25 mg SC QOD	**Flu-like symptoms** (premedicate acetaminophen/NSAID, dose at bedtime), **injection-site reactions/necrosis**, **depression**, ↑ LFTs, cytopenias, thyroid dysfunction • neutralizing antibodies
Glatiramer acetate	20 mg SC daily or 40 mg SC 3×/week	Injection-site reactions, **lipoatrophy**, **immediate post-injection reaction** (flushing, chest tightness, palpitations, dyspnea — transient), rare anaphylaxis • considered safe in pregnancy
Teriflunomide	7–14 mg daily	**Boxed: hepatotoxicity, teratogenicity** (men and women need contraception) • hair thinning, diarrhea, ↑ BP • **accelerated elimination** with **cholestyramine** or activated charcoal
Dimethyl fumarate / diroximel fumarate	DMF 120 mg BID × 7 days → **240 mg BID** • diroximel 231 mg BID → 462 mg BID	**Flushing** (take **with food**; **aspirin 325 mg** 30 min before), **GI upset**, **lymphopenia** (CBC), **PML** (rare) • diroximel: fewer GI effects
S1P modulators: **fingolimod**, siponimod, ozanimod, ponesimod	Fingolimod 0.5 mg daily • others titrated	**First-dose bradycardia/AV block** (fingolimod: **6-hour monitoring**, ECG) • **macular edema** (eye exam) • infections (**VZV**), ↑ LFTs, ↑ BP, PRES • **rebound** after stopping • siponimod: **CYP2C9 genotype** (CI in *3/*3)
Cladribine tablets	Two yearly short courses (weight-based)	**Boxed: malignancy, teratogenicity** • lymphopenia, herpes zoster
Natalizumab (α4-integrin)	300 mg IV every 4 weeks	**Boxed: PML** (JC virus) → **JCV antibody** testing, restricted program • monotherapy only • infusion reactions, hepatotoxicity

DMT	Dose	Key ADRs / monitoring
Anti-CD20: **ocrelizumab** (also **PPMS**), ofatumumab, ublituximab	Ocrelizumab 300 mg IV × 2 (2 weeks apart) → **600 mg every 6 months** (SC form available) • ofatumumab 20 mg SC weeks 0, 1, 2, then monthly	**Infusion reactions** (premedicate methylprednisolone + antihistamine ± acetaminophen) • infections • **HBV reactivation** (screen) • ↓ immunoglobulins • vaccines ≥ 2–6 weeks before
Alemtuzumab	2 annual IV courses	**Boxed: autoimmune disorders** (ITP, anti-GBM disease, **thyroid**), **infusion reactions**, **stroke**, malignancy • REMS with monthly labs for 4 years
Mitoxantrone	Rarely used	**Cardiotoxicity** (lifetime limit), secondary AML

MS and pregnancy

- Relapses ↓ during pregnancy (especially 3rd trimester) and ↑ **postpartum**.
- Most DMTs are stopped before conception • **teratogenic**: **teriflunomide** (accelerated elimination), **fingolimod/S1P** (stop ~2 months before), **cladribine**, mitoxantrone • **glatiramer** and **interferon** may be continued • anti-CD20 timing planned before conception.

Symptomatic treatment

Symptom	Treatment
Walking	**Dalfampridine** 10 mg BID (K-channel blocker) • **CI: seizure history**, **CrCl ≤ 50**
Spasticity	**Baclofen** (start 5 mg TID; sedation; **do not stop abruptly** → seizures, hallucinations; intrathecal pump) • **tizanidine** (2–4 mg; hypotension, ↑ LFTs, sedation; **CI with ciprofloxacin and fluvoxamine** — CYP1A2) • dantrolene (hepatotoxicity) • botulinum toxin • gabapentin
Fatigue	Amantadine, modafinil, methylphenidate, exercise, cooling
Bladder	Overactive: oxybutynin, tolterodine, **mirabegron** • retention: catheterization
Depression	SSRIs, SNRIs, CBT
Neuropathic pain	Gabapentin, pregabalin, duloxetine, TCAs
Pseudobulbar affect	**Dextromethorphan/quinidine** (Nuedexta) — QT
General	Avoid heat • **vitamin D** • exercise • smoking cessation • vaccinations (inactivated)

MS drug summary

Drug	Dose	Key ADRs	Key DDIs / notes
Methylprednisolone IV	1 g daily × 3–5 days	Hyperglycemia, insomnia, mood, GI upset	Relapse only
Interferon beta	See DMT table	**Flu-like**, depression, ↑ LFTs, injection sites	CBC, LFTs, TSH
Glatiramer	20 mg daily or 40 mg TIW SC	Post-injection reaction, lipoatrophy	Pregnancy option
Teriflunomide	7–14 mg daily	**Hepatotoxicity, teratogen**, hair thinning	Cholestyramine washout • ↑ warfarin, rosuvastatin
Dimethyl fumarate	240 mg BID	**Flushing**, GI, lymphopenia	Food + aspirin for flushing
Fingolimod	0.5 mg daily	**Bradycardia**, macular edema, infections	First-dose 6-h monitoring • avoid class Ia/III antiarrhythmics, beta-blockers/CCBs caution
Siponimod / ozanimod / ponesimod	Titrated daily	Same class effects	Siponimod CYP2C9 genotype • ozanimod CI with MAOIs
Cladribine	Weight-based short courses	**Malignancy, teratogen**, lymphopenia	—
Natalizumab	300 mg IV q4 wk	**PML**	JCV testing • no other immunosuppressants
Ocrelizumab	600 mg IV q6 months (after 2 × 300 mg)	Infusion reactions, infections, HBV reactivation	Premedicate • PPMS approved
Ofatumumab	20 mg SC weeks 0, 1, 2 → monthly	Injection reactions, infections	HBV screen
Alemtuzumab	IV yearly × 2 courses	**Autoimmunity**, infusion reactions, stroke	REMS labs × 4 years
Dalfampridine	10 mg BID (12 h apart)	**Seizures**, UTI, insomnia	CI seizures, CrCl ≤ 50
Baclofen	5 mg TID → max 80 mg/day	Sedation, weakness	**Taper** (withdrawal seizures) • renal adjustment
Tizanidine	2–4 mg → max 36 mg/day	Hypotension, sedation, ↑ LFTs	**CI with ciprofloxacin, fluvoxamine** (CYP1A2) • smoking ↓ levels
Dextromethorphan/ quinidine	20/10 mg daily → BID	Dizziness, diarrhea, QT	CYP2D6 inhibition (quinidine)

6. Dementia (Alzheimer's Disease)

Overview

- **Dementia:** progressive cognitive decline (memory + another domain) that interferes with daily function.
- **Types: Alzheimer's disease (most common)** • **vascular** • **Lewy body** (visual hallucinations, parkinsonism, fluctuations; **severe sensitivity to antipsychotics**) • frontotemporal (personality, behavior changes) • Parkinson's disease dementia.
- **Alzheimer's pathology: β-amyloid plaques** + **tau neurofibrillary tangles** → loss of **cholinergic** neurons.
- **Assessment: MMSE** (≥ 25 normal; 20–24 mild; 13–20 moderate; ≤ 12 severe) • **MoCA** (< 26 abnormal).
- **Rule out reversible causes: B12 deficiency**, **hypothyroidism**, depression ("pseudodementia"), **drugs**, infection, normal pressure hydrocephalus, electrolyte problems.

Drugs that worsen cognition (deprescribe)

Anticholinergics (diphenhydramine, hydroxyzine, oxybutynin, **TCAs**, paroxetine, benztropine) • **benzodiazepines**, **Z-drugs** • **opioids** • antipsychotics • cimetidine • muscle relaxants. **Anticholinergics also cancel the effect of cholinesterase inhibitors.**

Cholinesterase inhibitors (mild, moderate, and severe AD)

Drug	Dose	Notes
Donepezil	**5 mg at bedtime** × 4–6 weeks → **10 mg** (23 mg for moderate–severe)	Once daily • if vivid dreams/insomnia → give in the morning • mild to severe
Rivastigmine	Oral 1.5 mg BID **with food** → max 6 mg BID • **patch** 4.6 → 9.5 → 13.3 mg/24 h	Patch has **fewer GI effects** • **restart titration** if missed > 3 days • also **Parkinson's disease dementia** • not hepatically metabolized
Galantamine	4 mg BID → 12 mg BID (ER once daily)	Renal/hepatic dose limits • mild–moderate only

- **ADRs (cholinergic): nausea, vomiting, diarrhea**, **anorexia/weight loss**, **bradycardia** and **syncope** (caution: heart block, sick sinus), vivid dreams/insomnia, urinary incontinence, ↑ gastric acid (GI bleed risk with NSAIDs), bronchospasm (caution asthma/COPD), muscle cramps.
- **DDIs: anticholinergics** oppose effect • **beta-blockers**, digoxin, verapamil → additive bradycardia • prolong **succinylcholine** effect.
- Expect **modest** benefit (slower decline) • assess after 3–6 months • titrate slowly to reduce GI effects.

Memantine (moderate–severe AD)

- **NMDA receptor antagonist** → ↓ glutamate excitotoxicity.
- **Dose: 5 mg daily**, ↑ by 5 mg/week to **10 mg BID** (ER: 7 → 28 mg once daily) • **CrCl < 30: 5 mg BID**.
- **ADRs:** dizziness, confusion, headache, **constipation**, agitation (generally well tolerated).
- Often **combined with donepezil** (Namzaric) in moderate–severe disease • alkaline urine ↓ clearance.

Anti-amyloid monoclonal antibodies (early AD)

Drug	Dose	Key points
Lecanemab (Leqembi)	10 mg/kg IV every 2 weeks (SC maintenance option)	For **mild cognitive impairment or mild dementia** with **confirmed amyloid** (PET or CSF)
Donanemab (Kisunla)	700 mg IV q4 wk × 3 → 1,400 mg q4 wk	Can **stop when amyloid is cleared** on PET

- **Boxed warning: ARIA** (amyloid-related imaging abnormalities — **brain edema** and **microhemorrhages**) → **baseline and periodic MRI** • highest risk in **ApoE ε4 homozygotes** (test before starting).
- Caution with **anticoagulants and thrombolytics** (brain hemorrhage) • infusion reactions • only slows decline modestly.

Behavioral and psychological symptoms (BPSD)

- **Non-drug approaches first** (routine, environment, treat **pain**, infection, constipation, hearing/vision).
- **Antipsychotics** only for **severe agitation/psychosis** with danger → lowest dose, shortest time • **boxed warning: ↑ mortality in elderly patients with dementia** (stroke, infection) • **brexpiprazole** is FDA-approved for **agitation in Alzheimer's dementia** • **avoid** typical antipsychotics in **Lewy body dementia** (severe sensitivity).
- **Depression:** SSRIs (sertraline, citalopram ≤ 20 mg if > 60, escitalopram) • avoid TCAs and paroxetine.
- **Insomnia:** sleep hygiene • **suvorexant** (approved for insomnia in AD) • low-dose doxepin, melatonin • **avoid benzodiazepines, Z-drugs, diphenhydramine**.

Dementia drug summary

Drug	Dose	Key ADRs	Key DDIs / notes
Donepezil	5 mg HS → 10 mg (23 mg)	N/V/D, **bradycardia**, syncope, vivid dreams	Anticholinergics oppose • beta-blockers additive • CYP2D6/3A4
Rivastigmine	1.5 mg BID → 6 mg BID with food • patch 4.6–13.3 mg/24 h	GI (worse orally), bradycardia, application-site	Re-titrate after > 3 days missed • rotate patch sites
Galantamine	4 mg BID → 12 mg BID	GI, bradycardia, rare SJS	Renal/hepatic limits • CYP2D6/3A4
Memantine	5 mg daily → 10 mg BID (ER 28 mg daily)	Dizziness, confusion, constipation	CrCl < 30: 5 mg BID • alkaline urine ↓ clearance
Donepezil/memantine (Namzaric)	7/28 → 10/28 mg daily (ER)	Combined	Moderate–severe AD
Lecanemab	10 mg/kg IV q2 wk	**Boxed: ARIA**, infusion reactions	MRI monitoring • ApoE ε4 testing • caution anticoagulants
Donanemab	700 mg q4 wk × 3 → 1,400 mg q4 wk	**Boxed: ARIA**, infusion reactions	Same
Brexpiprazole	0.5 mg → 2–3 mg daily (AD agitation)	Weight gain, akathisia	**Boxed: ↑ mortality in elderly dementia** • CYP2D6/3A4
Suvorexant	10–20 mg HS	Somnolence, next-day impairment	CYP3A4 inhibitors ↑

Exam Traps and Frequently Tested Points

Epilepsy

1. **Absence → ethosuximide** (or valproate) • **avoid carbamazepine, oxcarbazepine, phenytoin, gabapentin** in absence/myoclonic.
2. **Valproate**: most effective for generalized seizures but **avoid in women of childbearing potential** (neural tube defects, ↓ IQ).
3. **Lamotrigine**: slow titration (SJS) • **halve dose with valproate • estrogen ↓ levels**.
4. **Carbamazepine**: HLA-B*1502 → SJS • autoinduction • SIADH • aplastic anemia.
5. **Phenytoin**: non-linear kinetics, **albumin correction**, gingival hyperplasia • IV max 50 mg/min.
6. **Levetiracetam**: behavior changes; renal dosing; few DDIs.
7. **Topiramate**: kidney stones, metabolic acidosis, glaucoma, cognitive slowing.
8. **Carbapenems ↓ valproate**.
9. **Enzyme inducers ↓ oral contraceptives** → IUD/depot.
10. **Status epilepticus: benzodiazepine first** (lorazepam IV, midazolam IM) → fosphenytoin/valproate/levetiracetam → anesthetic infusion.
11. Isoniazid seizures → **pyridoxine**.

Stroke

1. **Non-contrast CT first** to rule out hemorrhage.
2. **Alteplase 0.9 mg/kg (max 90 mg), 10% bolus** • tenecteplase 0.25 mg/kg bolus (max 25 mg) • within **3–4.5 h**.
3. **BP < 185/110 before**, **< 180/105 for 24 h after • no antithrombotics for 24 h**.
4. **No tPA:** permissive HTN up to 220/120 • **aspirin within 24–48 h**.
5. **Minor stroke/high-risk TIA → aspirin + clopidogrel × 21 days**.
6. **AF → DOAC** (not aspirin) • **high-intensity statin** • BP < 130/80.
7. **SAH → nimodipine 60 mg q4h × 21 days** (never IV).

Parkinson's disease

1. **TRAP • carbidopa/levodopa** most effective • carbidopa prevents peripheral conversion.

2. **Dopamine agonists** → sleep attacks, **impulse-control disorders**, hallucinations, edema.
3. **Selegiline** → insomnia • **MAO-B inhibitors + meperidine** → contraindicated.
4. **Entacapone** → orange urine, diarrhea • **tolcapone** → hepatotoxicity.
5. **Amantadine** → dyskinesia, livedo reticularis.
6. **Anticholinergics** → tremor in young only; avoid in elderly.
7. **Apomorphine + ondansetron** → contraindicated (hypotension).
8. **PD psychosis → pimavanserin**, quetiapine, clozapine • avoid haloperidol, risperidone, metoclopramide.
9. **Rivastigmine** → PD dementia.

Headache

1. **Triptans CI:** CAD, stroke, uncontrolled HTN, hemiplegic migraine, MAOIs, ergots within 24 h.
2. **Rizatriptan 5 mg with propranolol • eletriptan + strong CYP3A4 inhibitors** CI.
3. **Gepants and lasmiditan** → safe with CV disease • lasmiditan: no driving 8 h.
4. **MOH:** limit acute therapy to ≤ 2–3 days/week; triptans/opioids ≥ 10 days/month.
5. **Prevention** if ≥ 4 headache days/month • propranolol, topiramate, CGRP mAbs, gepants • botulinum toxin for chronic migraine.
6. **Erenumab** → constipation, hypertension.
7. **Migraine with aura → no estrogen contraceptives**.
8. **Cluster:** 100% O2 + SC sumatriptan • **verapamil** prevention (ECG).

Multiple sclerosis

1. **Relapse → methylprednisolone 1 g IV × 3–5 days**.
2. **Interferons** → flu-like symptoms, depression • **glatiramer** → post-injection reaction, lipoatrophy.
3. **Teriflunomide** → hepatotoxicity, teratogenic → cholestyramine washout.
4. **Dimethyl fumarate** → flushing (aspirin), lymphopenia.
5. **Fingolimod** → first-dose bradycardia (6-h monitoring), macular edema.
6. **Natalizumab** → **PML** (JCV antibodies).
7. **Ocrelizumab** → only DMT for **PPMS** • HBV screen.
8. **Alemtuzumab** → autoimmunity (thyroid, ITP).
9. **Dalfampridine** → seizures; CI CrCl ≤ 50.
10. **Tizanidine + ciprofloxacin/fluvoxamine** → contraindicated • **baclofen** → don't stop abruptly.

Dementia

1. **Rule out reversible causes** (B12, thyroid, depression, drugs).
2. **Cholinesterase inhibitors** → mild–severe AD • GI effects, **bradycardia/syncope**.
3. **Donepezil at bedtime** (morning if vivid dreams) • **rivastigmine with food**; patch fewer GI effects; re-titrate if missed > 3 days.
4. **Memantine** → moderate–severe; renal adjustment (CrCl < 30).
5. **Anticholinergics cancel cholinesterase inhibitors** → deprescribe (diphenhydramine, oxybutynin).
6. **Lecanemab/donanemab** → early AD with amyloid; **ARIA** (MRI, ApoE ε4).
7. **Antipsychotics in dementia** → boxed ↑ mortality; **brexpiprazole** approved for AD agitation.
8. **Lewy body dementia** → avoid typical antipsychotics.

> **Note:** Doses and guideline recommendations differ between references and are updated often. **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!